SOIL EROSION AND CONSERVATION · NOTE ER01

Land Degradation in India — Soil Erosion Extent, History, Causes and Conservation Efforts

Course topics: CAE1.3 L9, SWCE-501 T3, SWCE-501 T6, SWCE-505 T2, SWCE-505 T3, SWCE8.15 L11

What soil erosion is

Soil erosion is the detachment and transport of soil from one place to another by wind, moving water and the beating action of raindrops.

- Soil is the uppermost layer of the earth's crust — weathered and disintegrated minerals plus organic matter. Its depth varies from nil in some places to several metres in others.
- The **top 30 cm** supports human and animal life and gives plants physical support and nutrition. Conserving this thin layer is essential for mankind's survival.

Land and water degradation in India

- Land and water provide food, fodder, fibre, shelter and communication, and must support a growing population at a better standard of living.
- Of India's **329 million ha** geographical area, **173.64 million ha (≈ 53%)** faces degradation — from water and wind erosion, waterlogging, salinity and alkalinity, ravines and shifting cultivation. More frequent droughts and floods cut productivity further.
- **Magnitude of erosion:** about 5500 million tonnes of soil lost every year — an average of ≈ 16 t/ha/year, double the permissible limit.

Dated figures. The widely cited estimate is 5,334 million t/year, or 16.35 t/ha/year (Dhruva Narayana and Ram Babu, 1983). ISRO-SAC's Desertification and Land Degradation Atlas (2021) maps 97.85 million ha (about 29.7% of the country) as degraded in 2018–19, using a different method from the 173.64 million ha estimate. Quote the year with any figure.

Cause (degraded area)	Million ha
Water and wind erosion	144.42
Waterlogging	8.73
Alkali soils	3.59
Saline soils	5.50
Ravines and gullies	3.97
Shifting cultivation	4.91
Riverine and torrents	2.73
Total degraded area	**173.64**

Interlinked causes of soil loss: deforestation; shrinking natural pastures and overgrazing; erosion; farming unsuitable land; waterlogging; salt and alkali build-up; rising flood-proneness; loss of organic

matter and micronutrients; excessive farm chemicals; continuous monoculture; and poor water management — all rooted in disregard for nature and the land's capability.

History

- Erosion has followed the cultivation of sloping land everywhere and contributed to the decline of ancient civilisations.
- **Mesopotamia:** once fertile, now desert from erosion and floods. **Syria:** 1–2 m of topsoil stripped from hill slopes. **France:** forest clearing caused floods; forestry laws of 1860 and 1864 revegetated the mountainsides. **China:** large areas destroyed and abandoned — soil loss up to 207 t/ha on a 15% slope.
- **India:** the decline of Mohenjodaro–Harappa is linked to land neglect, and the Saraswati river vanished as its watershed was over-exploited. Overgrazing in the Siwalik hills caused erosion and siltation; the **Chos Act (1900)** controlled grazing in Punjab.
- In India erosion shows as fissures, gullies and ravines along rivers — the Chambal is worst affected. Traditional bunds and tanks control runoff but cannot fully check erosion.

Global land-use picture

- About 430 million ha worldwide is affected by erosion.
- Soils degrade by loss of minerals and organic matter; salinity, waterlogging and chemical poisoning; physical loss by water and wind; and conversion of farmland to reservoirs, buildings and roads — leaving less capacity to feed a growing population.
- **Land system:** geology, topography, hydrology, soils, micro-organisms, vegetation, animals, climate and people interact; land-use decisions depend on physical factors, technology and socio-economic and political considerations.
- **Land Degradation Neutrality (LDN)** aims at balanced use of land resources. It emerged from Rio+20 (2012) and was adopted at UNCCD COP-12 (2015).

Floods, droughts and siltation

- Flood-prone area ≈ 40 million ha, of which ≈ 8 million ha is affected every year; drought-prone area ≈ 26 million ha.
- **Siltation** is the deposition of water-borne sediment in rivers, lakes, canals and reservoirs when the flow slows down. Sediment rates observed in reservoirs are far above the rates assumed in design, so their useful life drops drastically. Silted rivers carry less, causing floods and damage — so the catchments need soil conservation.

Erosion status and conservation efforts in India

- Gully heads are advancing in Jharkhand; 70% of cultivated land is affected in parts of Maharashtra and Gujarat; black soils at Sholapur lose up to 300 t/ha/year. The main causes are excessive deforestation, overgrazing and faulty farming, which strip fertile topsoil and cut yields.

Period	Effort
After Independence	Soil and Water Conservation Division set up
2nd Five-Year Plan	All India Soil and Land Use Survey (AISLUS) and research centres
3rd Plan	Catchment soil-conservation scheme (expanded later)
4th Plan onward	Integrated watershed approach
5th Plan	Alkali soil reclamation and control of shifting cultivation
6th Plan	Watershed management in flood-prone rivers
7th Plan	Ravine reclamation projects
Throughout	National and State Land Use Boards; Central Soil and Water Conservation Research and Training Institute, Dehradun, for research and training

- **Gaps:** technology does not reach farmers well, and there is too little farmer interaction, integration with crop programmes, engineering planning and erosion-proof outlets — so gullies form and land degrades.
- **Present status:** forests and grasslands are degrading fast. India has 15% of the world's population but only 2% of its forest area; marginal land should shift to permanent vegetation.

Updated: India now has about 18% of the world's population (UN World Population Prospects 2024); 15% is an older figure.

Soil conservation regions

India is divided into **10 soil conservation regions** by soil, rainfall, physiography, vegetation and the erosion and degradation problems. Common regional problems are low productivity, low employment, low per-capita income and fragmented holdings that make conservation difficult.

Soil group	Area	Rainfall	Problems
Black soils (Vertisols)	≈ 70 million ha	500–1500 mm	Runoff up to 40%; loss of depth lowers productivity, raises drought-proneness and stalls yields
Red soils	≈ 72 million ha	750–2000 mm	Generally shallow, low water retention, prone to erosion and runoff

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. ISRO Space Applications Centre, Desertification and Land Degradation Atlas of India (2021). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. India's degraded area (older national estimate) is about

(a) 97 Mha (b) 144.42 Mha
(c) 173.64 Mha (d) 329 Mha

Q2. Area degraded by water and wind erosion in India is about

(a) 8.73 Mha (b) 44.4 Mha
(c) 144.42 Mha (d) 173.64 Mha

Q3. Average soil loss in India is about

(a) 1.6 t/ha/yr
(b) 16 t/ha/yr
(c) 160 t/ha/yr
(d) 300 t/ha/yr

Q4. Land Degradation Neutrality was adopted at

(a) Rio 1992
(b) Kyoto 1997
(c) UNCCD COP-12, 2015
(d) COP-21 Paris

Q5. The Chos Act (1900) aimed to control

(a) canal seepage
(b) grazing in Punjab
(c) shifting cultivation
(d) mining

Q6. Ravine reclamation projects were taken up in the

(a) 3rd Plan
(b) 5th Plan
(c) 7th Plan
(d) 10th Plan

Q7. India is divided into how many soil conservation regions?

(a) 5
(b) 8
(c) 10
(d) 15

Answers

Q1 — (c) 173.64 Mha — ≈ *53% of 329 Mha.*

Q2 — (c) 144.42 Mha

Q3 — (b) 16 t/ha/yr — *About double the permissible limit.*

Q4 — (c) UNCCD COP-12, 2015

Q5 — (b) grazing in Punjab

Q6 — (c) 7th Plan

Q7 — (c) 10